

## Training Configuration and Summary:

### Sequencing:

- timesteps: 30
- batch\_size: 128

### Model Hyperparameters:

- min\_units: 128
- max\_units: 256
- units\_step: 64
- dropout\_min: 0.2
- dropout\_max: 0.3
- dropout\_step: 0.1
- activation\_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

### Tuning:

- max\_epochs: 30
- hyperband\_iterations: 3

## LSTM Multi-Step Training Report

Report Generated: 2025-04-14 23:07:04

Training Start: 2025-04-14\_19-48-56

Training End: 2025-04-14\_20-27-44

Training Duration: 38.79 minutes

### Best Hyperparameters:

Number of Layers: 2

Activation: relu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 1.4255

MSE: 3.9646

RMSE: 1.9911

R2 Score: 0.9100

### Multi-Step Forecast:

Forecast Steps: 5

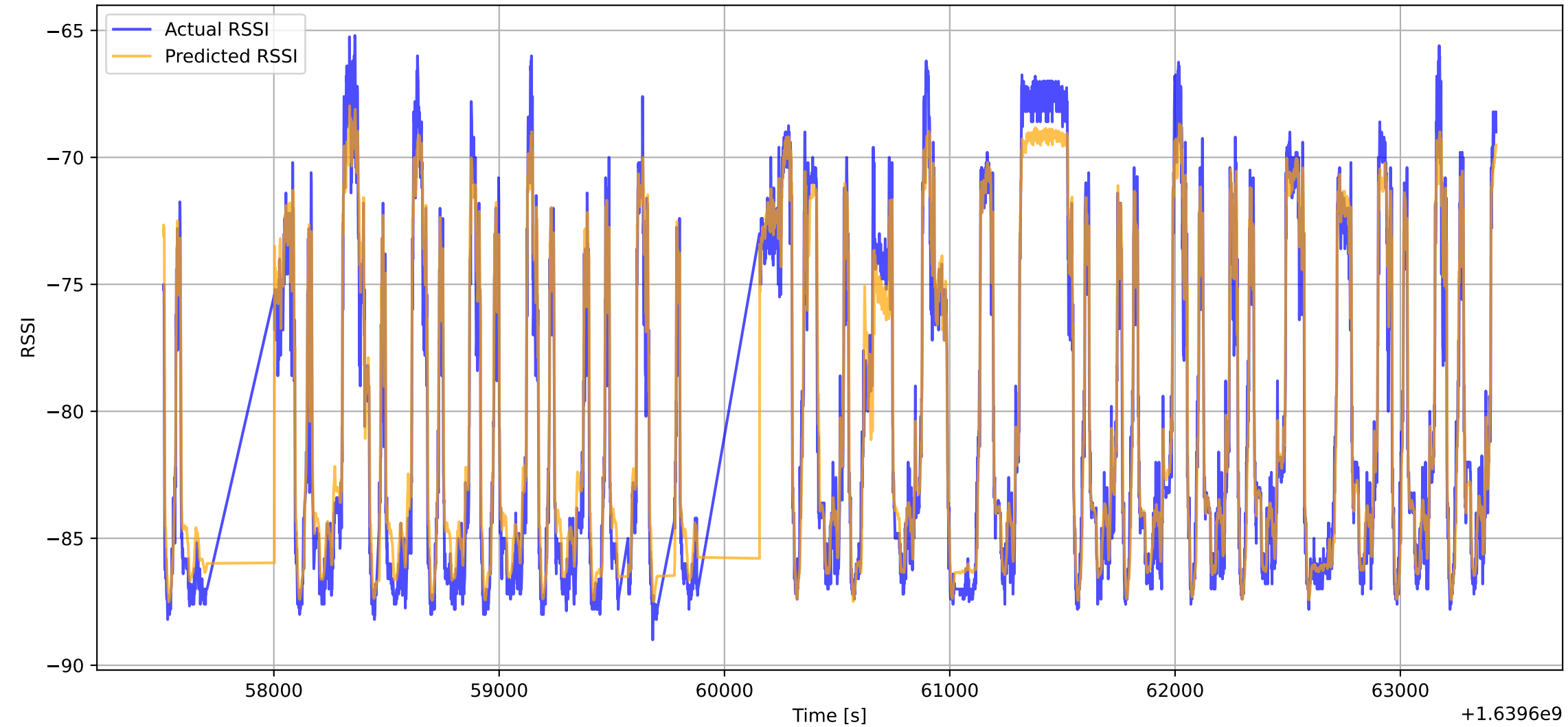
MAE: 0.0310

MSE: 0.0017

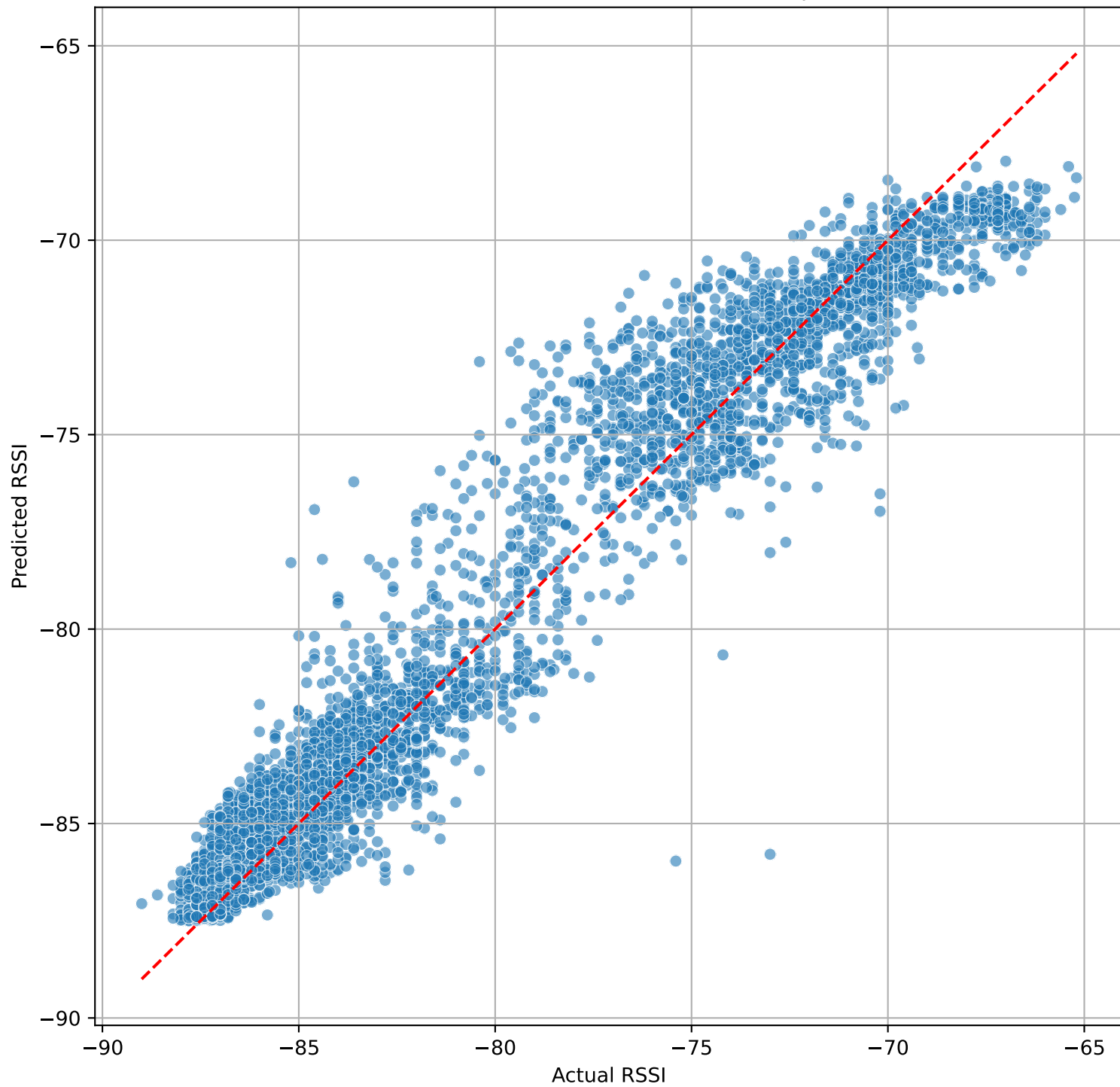
RMSE: 0.0415

R2 Score: 0.9498

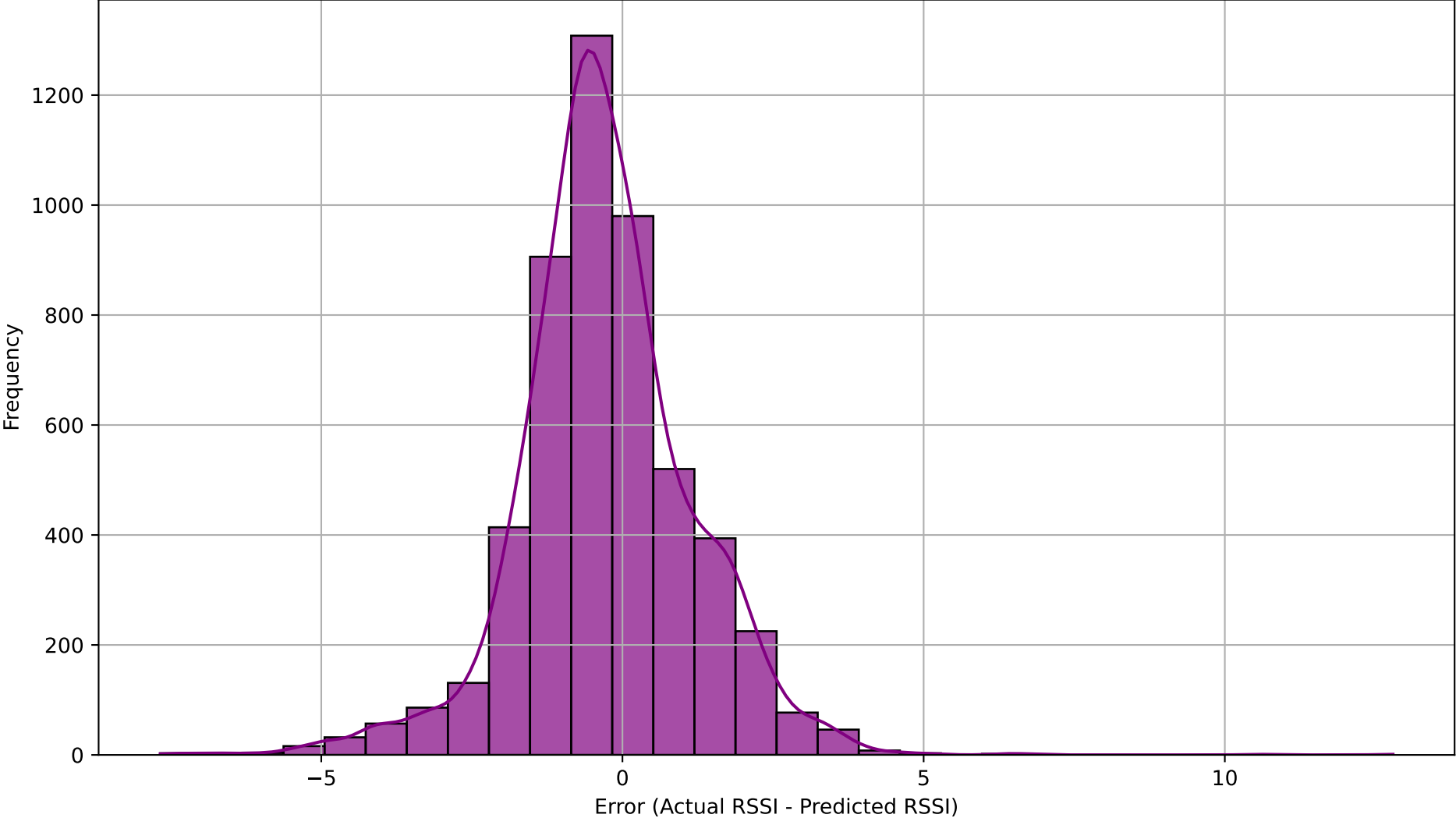
Actual vs Predicted RSSI (First Step)



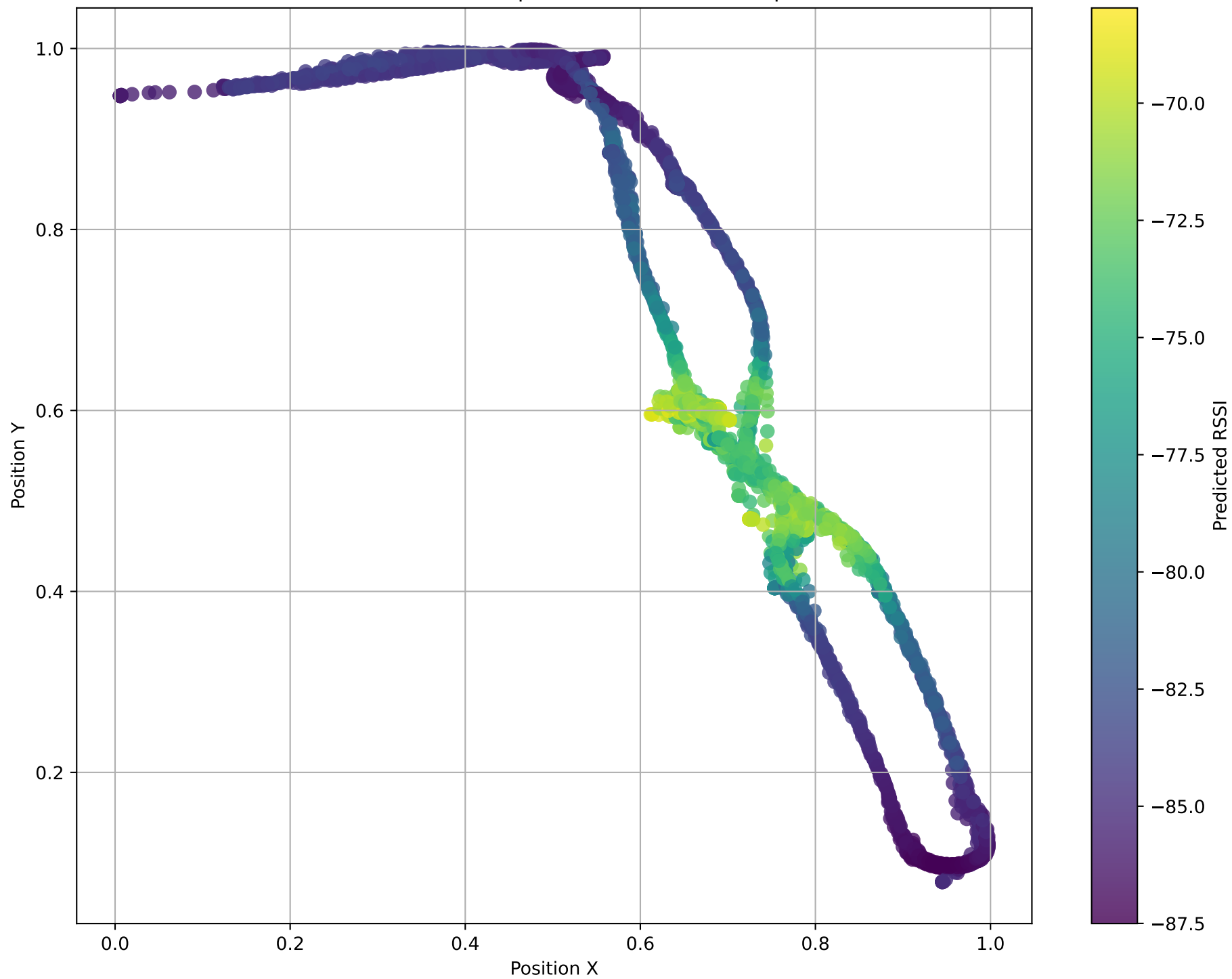
Actual vs Predicted RSSI (First Step)



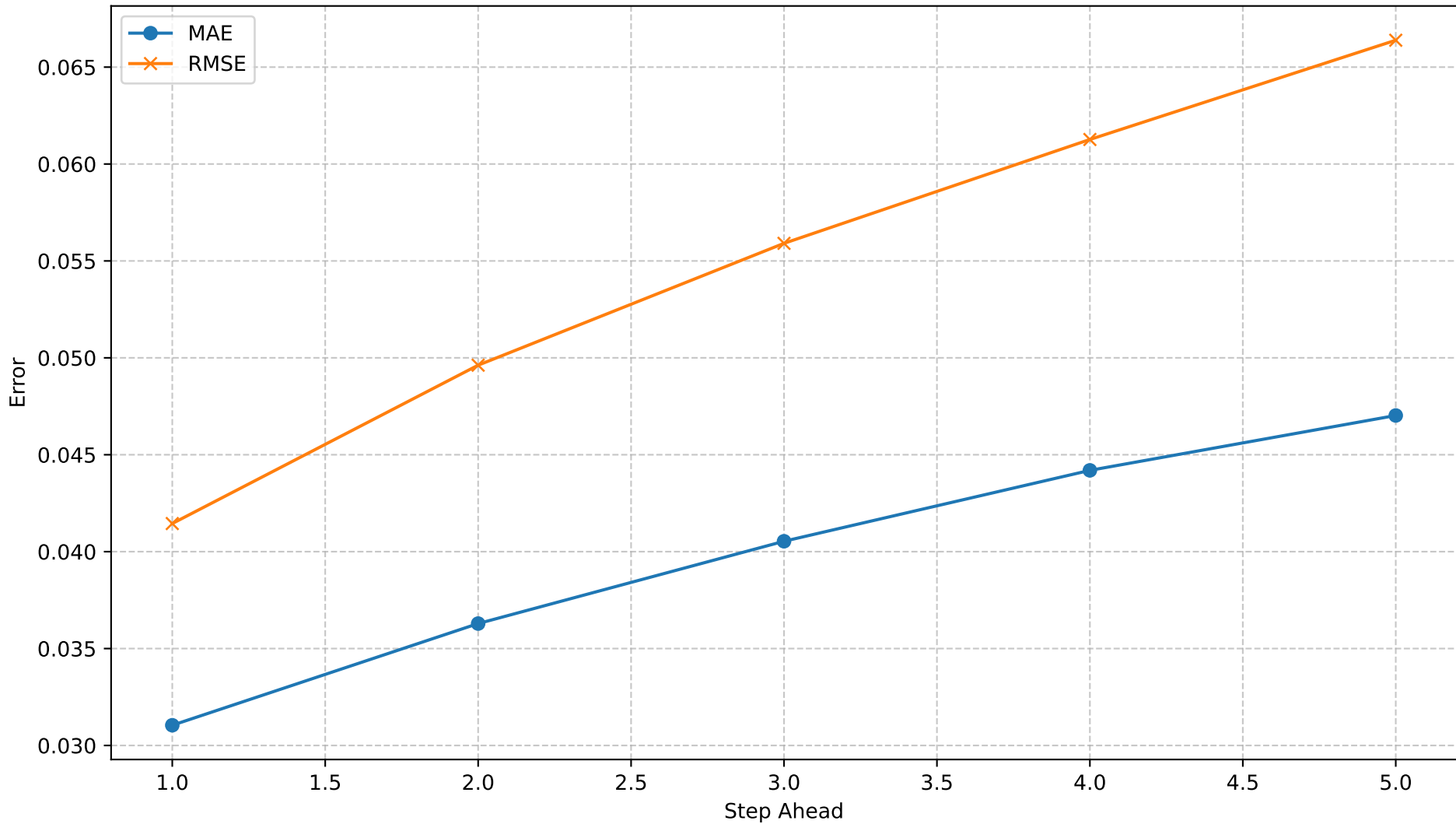
Distribution of Prediction Errors (First Step)



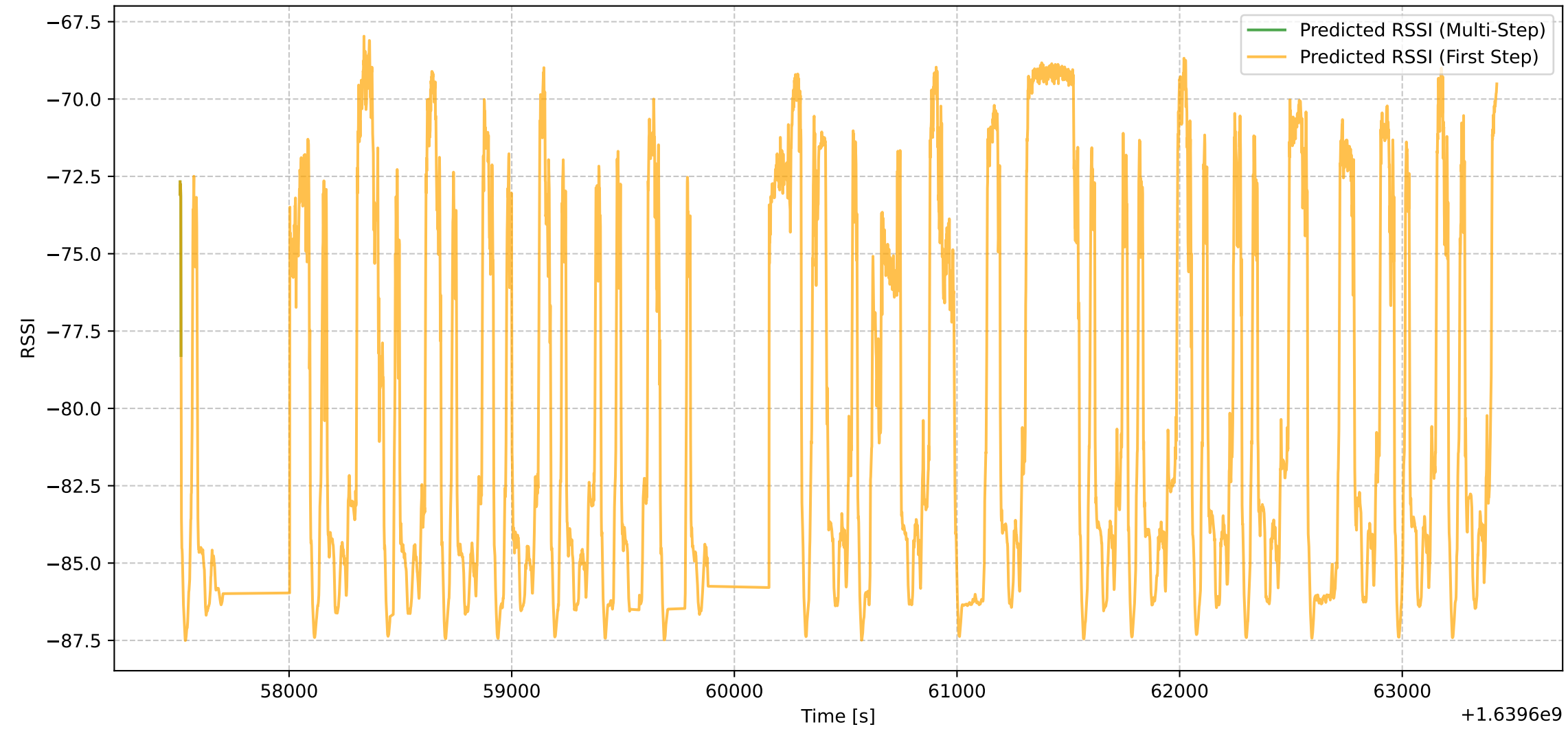
Predicted RSSI in Spatial Context (First Step)



Multi-Step Forecast Error by Step



Actual vs Predicted RSSI (First Step vs Multi-Step)



#### Resumo Automático:

- Forecast Steps configurados: 5
- Passos com MAE  $\leq 2.0$ : 5 passo(s)
- MAE no último passo: 0.0470

□ O modelo mantém boa acurácia durante quase toda a sequência de previsões.